Appendix A Final Mc RST Scale

Metacognition about Regulating Recreational Screen Time Scale

The final scale contains 13 items: metacognitive awareness (MA; 5 items), planning (3), monitoring (2), and evaluation (3). The Chinese items reproduce the administered simplified-Chinese questionnaire, retaining only the final items. The English items reproduce the manuscript wording and are provided for reference; an English-language version has not been independently validated.

Administration

The questionnaire asks about general beliefs and usual habits. No fixed retrospective recall period was specified. Administer the MA items with agreement anchors and the regulation items with frequency anchors; retain these distinct response formats.

Chinese instructions

元认知意识：以下题目旨在了解您对调节娱乐屏幕时间的想法与信念。

娱乐屏幕时间是指您在非工作或学业需求的情况下，使用电子设备（如手机、电脑、平板、电视等）进行娱乐活动的时间，例如浏览社交媒体、观看视频、玩电子游戏等。

请根据您的真实想法，评估以下陈述，并选择最符合您的选项。

元认知调节：以下题目旨在了解您如何调节娱乐屏幕时间。

请根据您的真实使用习惯，评估以下陈述，并选择最符合实际情况的选项。

English reference translation of instructions

Metacognitive Awareness (MA): The following items concern your thoughts and beliefs about regulating recreational screen time. Recreational screen time refers to using electronic devices (e.g., smartphones, computers, tablets, or televisions) for entertainment unrelated to work or study, such as browsing social media, watching videos, or playing video games. Please indicate the response that best reflects your actual views.

Metacognitive Regulation (MR; including planning; monitoring; evaluation): The following items concern how you regulate recreational screen time. Please indicate the response that best reflects your usual habits.

Response anchors

Score	MA Chinese / English	MR Chinese / English
1	完全不同意 / Strongly disagree	从不 / Never
2	比较不同意 / Somewhat disagree	很少 / Rarely
3	不确定 / Uncertain	有时候 / Sometimes
4	比较同意 / Somewhat agree	经常 / Often
5	完全同意 / Strongly agree	总是 / Always

Metacognitive awareness

Use the MA agreement anchors for all five items below.

Final label	Chinese wording	English reference wording
MA01	我相信自己能够有效地调节娱乐屏幕时间。	I believe that I can effectively regulate my recreational screen time.
MA02	我了解自己调节娱乐屏幕时间的能力。	I am aware of my ability to regulate my recreational screen time.
MA03	我知道可以采取不同的方法来调节娱乐屏幕时长。	I know that I can adopt different strategies to regulate my recreational screen time.
MA04	我有信心能借助以往的时间管理经验来调节娱乐屏幕时间。	I am confident that I can draw on my past time management experience to regulate my recreational screen time.
MA05	我知道如何有效地调节娱乐屏幕时间。	I know how to effectively regulate my recreational screen time.

Metacognitive regulation

Use the regulation frequency anchors for all eight items below.

Final label	Chinese wording	English reference wording
MR-P01	在开始娱乐屏幕活动之前，我通常会先想一想动机是什么（比如，减压、打发时间）。	Before engaging in recreational screen activities, I usually think about my motivation (e.g., to relax, to pass time).
MR-P02	我会在开始娱乐屏幕活动之前先考虑使用时长。	Before starting recreational screen activities, I consider how long I will engage in them.
MR-P03	我会规划何时开始以及如何分配娱乐屏幕时间。	I plan when to start and how to allocate my recreational screen time.
MR-M01	当我觉得娱乐屏幕时间过长时，我会马上停下来。	When I feel that my recreational screen time is too long, I stop immediately.
MR-M02	当我觉得娱乐屏幕时间过长时，我会想办法做出调整。	When I feel that my recreational screen time is too long, I try to make adjustments.
MR-E01	我会回顾娱乐屏幕时间的使用是否有助于自身的身心健康。	I review whether my recreational screen time contributes to my physical and mental well-being.
MR-E02	我会反思自己在调节娱乐屏幕时间方面的表现如何。	I reflect on how well I regulate my recreational screen time.
MR-E03	我会探索是否有更好的方法来调节娱乐屏幕时间。	I explore whether there are better ways to regulate my recreational screen time.

Scoring

Code each response from 1 to 5. No final item is reverse-scored. Calculate arithmetic means of the items listed below; all mean scores range from 1 to 5, with higher scores indicating higher self-reported skills. The scoring labels below refer to the final 13-item scale, not the archived analysis variable names.

Score	Items	Calculation
MA	MA01–MA05	Sum of 5 MA responses / 5
Planning	MR-P01–MR-P03	Sum of 3 planning responses / 3
Monitoring	MR-M01–MR-M02	Sum of 2 monitoring responses / 2
Evaluation	MR-E01–MR-E03	Sum of 3 evaluation responses / 3
MR composite	All 8 planning, monitoring, and evaluation items	Sum of 8 regulation responses / 8
Mc-RST total	All 13 items	Sum of all 13 responses / 13

The total is an item-weighted mean; it is not the unweighted average of MA and MR or of the four dimension means. Likewise, the MR composite averages eight items rather than weighting the three regulation dimensions equally. An equivalent sum score ranges from 13 to 65; state explicitly when sums are used. Composite scores are observed item summaries, not latent factor scores.

Use complete responses for these calculations. The development studies did not establish a prorating rule for missing items. No diagnostic cutoff is available. The instrument is intended for research and formative assessment.